

CSE 280 Challenge Set 11

First & Last Name: _____

Question 1

Use the Euclid Algorithm to find the $\gcd$ for the following pairs of numbers. Use tables as we did in class. Note that you may not need all the rows in the table below.

- $\gcd(80, 32)$

$x = r'$	$y = x'$	$r = y \bmod x$

- $\gcd(560, 792)$

$x = r'$	$y = x'$	$r = y \bmod x$

Question 2

Use the Extended Euclid Algorithm to express the answers to question 1 as a linear combination.

- $\gcd(80, 32) = s * 80 + t * 32$

$x = r'$	$y = x'$	$r = y \bmod x$	$q = y \div x$	$s = t' - qs'$	$t = s'$

• $gcd(560, 792) = s * 560 + t * 792$

$x = r'$	$y = x'$	$r = y \bmod x$	$q = y \operatorname{div} x$	$s = t' - qs'$	$t = s'$

Question 3

Part 1

Create public and private RSA keys. Use the two prime numbers: $p = 163$ and $q = 431$. Calculate N (public) and ϕ . Select the **smallest** value e (public) such than $gcd(e, \phi) = 1$. Use the Extended Eulicd Algorithm to find the multiplicative inverse of $e \bmod \phi$ and call the result d (private).

Determine the equations to encrypt and decrypt. Use the table below to execute the Euclid algorithm.

$x = r'$	$y = x'$	$r = y \bmod x$	$q = y \operatorname{div} x$	$s = t' - qs'$	$t = s'$

Part 2

Use a python terminal (which is good at working with large numbers) to encrypt and decrypt the word "PIE" using the RSA keys obtained in Part 1 above. To convert the letters to numbers, use the ASCII code provided in the second column. Use the Python terminal to do the math.

Letter	ASCII	Encrypted	Decrypted	Letter
P	80			
I	73			
E	69			